

STA 332/MATH 343: Homework 1 – Solutions

Duke University, Fall 2026

Due: September 6th, 11:59 PM on Gradescope

Theory

1. Briefly define the notion of statistic.

Given a statistical model $\{f_\theta(x) \mid \theta \in \Theta\}$ and a random sample $\underline{X} = (X_1, \dots, X_n)$, a statistic is a function $T := T(\underline{X})$ which depends only on the random sample and in particular does not depend on the parameter θ .

2. Consider the statistical model $\{\mathcal{N}(\mu, \sigma^2) \mid \mu \in \mathbb{R}\}$, with σ^2 known, and let $\underline{X} = (X_1, \dots, X_n)$ be a random sample. What is the parameter space? For each of the following functions write whether they are statistics or not and why.

$$T_1(\underline{X}) = \sum_{i=1}^n X_i^2, \quad T_2(\underline{X}) = X_3 + 1, \quad T_3(\underline{X}) = \sum_{i=1}^n X_i^4 + \sigma^2, \quad T_4(\underline{X}) = \mu e^{X_n}.$$

The parameter space is $\Theta = \mathbb{R}$. Then T_1 and T_2 are statistics because they depend solely on $\underline{X}$, T_3 is a statistic because it depends only on $\underline{X}$ and σ^2 , which is assumed to be known. T_4 instead is **not** a statistic, because it depends on the parameter μ .

3. Consider the statistical model $\{\mathcal{N}(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+\}$ and let $\underline{X} = (X_1, \dots, X_n)$ be a random sample. For each of the following functions write whether they are statistics or not and why.

$$T_1(\underline{X}) = \sum_{i=1}^n X_i^2, \quad T_2(\underline{X}) = X_3 + 1, \quad T_3(\underline{X}) = \sum_{i=1}^n X_i^4 + \sigma^2, \quad T_4(\underline{X}) = \mu e^{X_n}.$$

The parameter space is $\Theta = \mathbb{R} \times \mathbb{R}_+$. Then T_1 and T_2 are statistics because they depend solely on $\underline{X}$, while T_3 and T_4 are **not** statistics since they depend on parameters σ^2 and μ respectively.

Probability Recap

4. Let $X \sim \text{Poisson}(\lambda)$. Compute $\text{Var}(X)$. (*Hint: note that $x^2 = x(x+1-1) = x(x-1) + x$.*)

Using the definition of expected value and the hint we have that

$$\begin{aligned} \mathbb{E}[X^2] &= \sum_{x=0}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x e^{-\lambda}}{x!} + \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x e^{-\lambda}}{x!} + \mathbb{E}[X] = \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x e^{-\lambda}}{x!} + \lambda. \end{aligned}$$

Notice that when $x = 0$ and $x = 1$ we have that $x(x-1)\frac{\lambda^x e^{-\lambda}}{x!}$ is equal to zero. Thus we have

$$\begin{aligned}\sum_{x=0}^{\infty} x(x-1)\frac{\lambda^x e^{-\lambda}}{x!} &= \sum_{x=2}^{\infty} x(x-1)\frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=2}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-2)!} \\ &= \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2} e^{-\lambda}}{(x-2)!} = \lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y e^{-\lambda}}{(y)!} = \lambda^2,\end{aligned}$$

by taking the substitution $y = x - 2$. Thus we obtain

$$\mathbb{E}[X^2] = \lambda^2 + \lambda.$$

Therefore,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

5. Let $X \sim \text{Beta}(\alpha, \beta)$, whose density function is given by

$$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbb{1}_{(0,1)}(x),$$

where $\alpha, \beta > 0$ and $\Gamma(\cdot)$ is the Gamma function. Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

Let $f(x)$ be the pdf of $X \sim \text{Beta}(\alpha, \beta)$, and define $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$, then

$$\begin{aligned}f(x) &= \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbb{1}_{\{y \in (0,1)\}}. \\ \mathbb{E}[X] &= \int_0^1 x f(x) dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha+1-1} (1-x)^{\beta-1} dx.\end{aligned}$$

Observe that the integrand is the kernel of the distribution $\text{Beta}(\alpha + 1, \beta)$, therefore

$$\mathbb{E}[X] = \frac{1}{B(\alpha, \beta)} B(\alpha + 1, \beta) = \frac{\alpha}{\alpha + \beta}.$$

Now, $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, so

$$\mathbb{E}[X^2] = \int_0^1 x^2 f(x) dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha+2-1} (1-x)^{\beta-1} dx.$$

Again, observe that the integrand is the kernel of the distribution $\text{Beta}(\alpha + 2, \beta)$, therefore

$$\mathbb{E}[X^2] = \frac{1}{B(\alpha, \beta)} B(\alpha + 2, \beta) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}.$$

This implies that

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)} - \left(\frac{\alpha}{\alpha + \beta}\right)^2 \\ &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}\end{aligned}$$

6. Suppose X has density function $f_X(x)$ supported on $\mathbb{R}$.
- (a) Derive the density function of $Y = \mu + \sigma X$, where $\mu \in \mathbb{R}$ and $\sigma > 0$ are constants.
 - (b) Use this to prove that if $X \sim \mathcal{N}(0, 1)$, then $\mu + \sigma X \sim \mathcal{N}(\mu, \sigma^2)$.

The transformation $g(x) = \mu + \sigma x$ is monotone with inverse $g^{-1}(y) = (y - \mu)/\sigma$. Thus, by the theorem for the change of variables, we have that

$$f_Y(y) = f_X\left(\frac{y - \mu}{\sigma}\right) \cdot \frac{1}{\sigma}.$$

If $X \sim \mathcal{N}(0, 1)$, the density function of $Y = \mu + \sigma X$ is given by

$$f_Y(y) = f_X\left(\frac{y - \mu}{\sigma}\right) \cdot \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y - \mu)^2},$$

which is the density function of $\mathcal{N}(\mu, \sigma^2)$.

7. Let $X \sim \text{Gamma}(\alpha, \beta)$. Find the density function of $Y = \frac{1}{X}$.

Notice that the domain of Y is given by $\mathbb{R}_+$. Since the transformation is monotone with differentiable inverse, we could use the theorem for the change of variables.

We can also directly compute the distribution function. Let $y > 0$, then

$$\mathbb{P}(Y \leq y) = \mathbb{P}(1/X \leq y) = \mathbb{P}(X > 1/y) = 1 - \mathbb{P}(X \leq 1/y).$$

Differentiating with respect to y we get

$$f_Y(y) = f_X(1/y) \frac{1}{y^2} = \frac{1}{y^2} \frac{\beta^\alpha}{\Gamma(\alpha)} (1/y)^{\alpha-1} e^{-\frac{\beta}{y}} \mathbb{1}_{\{y>0\}} = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/y)^{\alpha+1} e^{-\frac{\beta}{y}} \mathbb{1}_{\{y>0\}}.$$

8. Let $X \sim \mathcal{N}(0, 1)$. Find the density function of $Y = |X|$ and compute $\mathbb{E}[Y]$. (*Hint: the transformation is not invertible*)

Notice that the domain of Y is given by $\mathbb{R}_+$. Let $y > 0$, then

$$\mathbb{P}(Y \leq y) = \mathbb{P}(|X| \leq y) = \mathbb{P}(-y \leq X \leq y) = F_X(y) - F_X(-y).$$

Differentiating with respect to y we get

$$f_Y(y) = f_X(y) + f_X(-y) = \sqrt{\frac{2}{\pi}} e^{-\frac{y^2}{2}} \mathbb{1}(y \geq 0).$$

Moreover, by definition of expected value we have that

$$\mathbb{E}[Y] = \int_0^\infty y \sqrt{\frac{2}{\pi}} e^{-\frac{y^2}{2}} dy.$$

With the substitution $z = y^2 \implies dz = 2y dy$ we obtain

$$\mathbb{E}[Y] = \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{z}{2}} dz = \frac{1}{\sqrt{2\pi}} \left[-2e^{-\frac{z}{2}} \right]_0^\infty = \sqrt{\frac{2}{\pi}}.$$

9. Consider the probability density function of a random variable X ,

$$f_X(x) = c|x|\mathbb{1}_{\{-2 \leq x \leq 2\}}.$$

- (a) Find the value of c . (*Hint: the density function needs to integrate to 1*).
 (b) Compute $\mathbb{P}(X > 1)$.

(a) To find the constant c we need to compute the integral of the density over the domain and impose it to be equal to 1. This leads to

$$\int_{-2}^2 f(x) dx = 2c \int_0^2 x dx = 4c = 1 \quad \Leftrightarrow \quad c = \frac{1}{4}$$

(b) Note that

$$\mathbb{P}(X > 1) = \int_1^2 \frac{1}{4}x dx = \frac{1}{8}x^2 \Big|_1^2 = \frac{3}{8}.$$

10. Let $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1)$ and let $Y = \min\{X_1, \dots, X_n\}$.

- (a) Find the density function of Y . (*Hint: try to compute $\mathbb{P}(Y \geq y)$*)
 (b) Compute $\mathbb{E}[Y]$.

Let $y \in (0, 1)$, which is the domain of Y . Then we have that

$$\begin{aligned} \mathbb{P}(Y > y) &= \mathbb{P}(\min\{X_1, \dots, X_n\} > y) = \mathbb{P}(X_1 > y, \dots, X_n > y) \\ &= \prod_{i=1}^n \mathbb{P}(X_i > y) \quad \text{as } X_i \text{ are independent} \\ &= \mathbb{P}(X_1 > y)^n \quad \text{as } X_i \text{ are identically distributed} \\ &= (1 - y)^n. \end{aligned}$$

Therefore,

$$\mathbb{P}(Y \leq y) = 1 - \mathbb{P}(Y > y) = 1 - (1 - y)^n.$$

Differentiating both sides with respect to y we obtain

$$f_Y(y) = n(1 - y)^{n-1} \mathbb{1}_{\{0 \leq y \leq 1\}}.$$

As regards the expected value:

$$\begin{aligned} \mathbb{E}[Y] &= \int_0^1 ny(1 - y)^{n-1} dy = \int_0^1 n(y - 1 + 1)(1 - y)^{n-1} dy \\ &= \int_0^1 n(1 - y)^{n-1} dy - \int_0^1 n(1 - y)^n dy = n \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \frac{1}{n+1}. \end{aligned}$$

Alternatively, we could note that $Y \sim \text{Beta}(1, n)$ and then apply exercise 2. above.

11. Consider the following bivariate density function of the vector (X, Y)

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{else} \end{cases}$$

- (a) Compute the marginal density functions $f_X(x)$ and $f_Y(y)$.
 (b) Compute $\mathbb{P}(X \leq \frac{1}{2} \mid Y = \frac{1}{2})$ and $\mathbb{P}(X \leq \frac{1}{2}, Y = \frac{1}{2})$.

(a)

$$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 (x + y) dy = x + \left[\frac{y^2}{2} \right]_0^1 = x + 1/2.$$

As f is symmetric in x and y , we have $f_Y(y) = y + 1/2$.

- (b) The conditional density of X given Y is given by

$$f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)} = \frac{x + y}{y + 1/2} \cdot \mathbb{1}_{\{0 \leq x \leq 1\}}.$$

Thus, conditionally on $Y = 1/2$ we have

$$f_{X|Y}(x \mid 1/2) = (x + 1/2) \cdot \mathbb{1}_{\{0 \leq x \leq 1\}}.$$

Therefore,

$$\begin{aligned} \mathbb{P}(X \leq 1/2 \mid Y = 1/2) &= \int_0^{1/2} f_{X|Y}(x \mid 1/2) dx = \int_0^{1/2} (x + 1/2) dx \\ &= \left(\frac{x^2}{2} + \frac{x}{2} \right) \Big|_0^{1/2} = 1/8 + 1/4 = 3/8. \end{aligned}$$

$$\mathbb{P}(X \leq 1/2, Y = 1/2) = \int_0^{1/2} \int_{1/2}^{1/2} f(x, y) dx dy = 0.$$

Indeed, the probability that X is equal to a specific value is 0, since it is a continuous random variable.

Ungraded bonus problems

12. Let $X \sim \text{Gamma}(\alpha, \beta)$ and let $\lambda \in \mathbb{R}$ be a constant.

- (a) Compute $\mathbb{E}[e^{\lambda X}]$. For which values of λ is the expected value finite?
 (b) Let $X \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$ and assume they are independent. Find $\mathbb{E}[e^{\lambda(X+Y)}]$ when it is finite.

(a) Note that

$$\mathbb{E}[e^{\lambda X}] = \int_0^\infty e^{\lambda x} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-(\beta-\lambda)x} dx$$

If $\lambda < \beta$, then we recognize the kernel of a Gamma distribution, with parameters α and $\beta - \lambda$. This means that

$$\mathbb{E}[e^{\lambda X}] = \frac{\beta^\alpha}{(\beta - \lambda)^\alpha} \int_0^\infty \frac{(\beta - \lambda)^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-(\beta-\lambda)x} dx = \left(\frac{\beta}{\beta - \lambda} \right)^\alpha.$$

If instead $\lambda \geq \beta$ we have that

$$\int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-(\beta-\lambda)x} dx = +\infty,$$

since

$$\int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-(\beta-\lambda)x} dx \leq \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} dx = +\infty,$$

for every $\alpha > 0$. Thus, when $\lambda < \beta$ the expected value is finite and is given by

$$\mathbb{E}[e^{\lambda X}] = \left(\frac{\beta}{\beta - \lambda} \right)^\alpha.$$

(b) Because X and Y are independent we have that

$$\mathbb{E}[e^{\lambda(X+Y)}] = \mathbb{E}[e^{\lambda X}] \mathbb{E}[e^{\lambda Y}].$$

Using the result from part (a) we obtain that when $\lambda < \beta$

$$\mathbb{E}[e^{\lambda(X+Y)}] = \left(\frac{\beta}{\beta - \lambda} \right)^{\alpha_1} \left(\frac{\beta}{\beta - \lambda} \right)^{\alpha_2} = \left(\frac{\beta}{\beta - \lambda} \right)^{\alpha_1 + \alpha_2}.$$

The result can also be obtained using the fact that $X+Y \sim \text{Ga}(\alpha_1 + \alpha_2, \beta)$, and then applying part (a).

13. Let $X_1 \sim \text{Poisson}(\lambda_1)$, $X_2 \sim \text{Poisson}(\lambda_2)$ be independent random variables.

(a) Prove that $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

(b) Prove that $X_1 \mid X_1 + X_2 = n \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$.

(a) By direct computation we have that for every integer $k \geq 0$,

$$\mathbb{P}(X_1 + X_2 = k) = \sum_{i=0}^k \mathbb{P}(X_1 = i, X_2 = k - i) = \sum_{i=0}^k \mathbb{P}(X_1 = i) \cdot \mathbb{P}(X_2 = k - i),$$

where the second equality follows from the independence of X_1 and X_2 . Now, using the definition of the Poisson distribution we have that

$$\begin{aligned} \mathbb{P}(X_1 + X_2 = k) &= \sum_{i=0}^k \frac{\lambda_1^i e^{-\lambda_1}}{i!} \frac{\lambda_2^{k-i} e^{-\lambda_2}}{(k-i)!} \\ &= e^{-(\lambda_1 + \lambda_2)} \sum_{i=0}^k \frac{\lambda_1^i \lambda_2^{k-i}}{i!(k-i)!} \\ &\stackrel{(i)}{=} e^{-(\lambda_1 + \lambda_2)} \frac{1}{k!} \sum_{i=0}^k \binom{k}{i} \lambda_1^i \lambda_2^{k-i} \end{aligned}$$

$$\stackrel{(ii)}{=} e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!},$$

where (i) is due to $\frac{1}{k!} \binom{k}{i} = \frac{1}{i!(k-i)!}$, and (ii) follows from the binomial theorem. By the definition of the Poisson distribution we conclude that $X_1 + X_2 \sim \text{Po}(\lambda_1 + \lambda_2)$.

(b) By the definition of conditional probability we have that for every integer $0 \leq k \leq n$,

$$\mathbb{P}(X_1 = k \mid X_1 + X_2 = n) = \frac{\mathbb{P}(X_1 = k, X_2 = n - k)}{\mathbb{P}(X_1 + X_2 = n)}.$$

By the independence of X_1 and X_2 , the numerator is equal to

$$\mathbb{P}(X_1 = k) \cdot \mathbb{P}(X_2 = n - k) = \frac{\lambda_1^k e^{-\lambda_1}}{k!} \cdot \frac{\lambda_2^{n-k} e^{-\lambda_2}}{(n-k)!}.$$

Using part (a), the denominator is equal to

$$\mathbb{P}(X_1 + X_2 = n) = \frac{(\lambda_1 + \lambda_2)^n e^{-(\lambda_1 + \lambda_2)}}{n!}.$$

Putting everything together we have that

$$\mathbb{P}(X_1 = k \mid X_1 + X_2 = n) = \frac{n!}{k!(n-k)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k},$$

which is the probability mass function of a $\text{Bin}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$ random variable.